

Q- Suppose you see the following MAC address table on a switch: Port 1 - 00:11:22:33:44:55, Port 2 - 66:77:88:99:AA:BB. If a frame arrives at the switch on Port 1 with destination MAC 66:77:88:99:AA:BB, explain step-by-step how the switch will forward this frame.

Port	MAC Address
Port 1	00:11:22:33:44:55
Port 2	66:77:88:99:AA:BB

A frame arrives on **Port 1** with destination MAC address:

66:77:88:99:AA:BB

Step 1: Frame arrives

The switch receives the Ethernet frame on **Port 1**.

Step 2: Check the destination MAC

The switch examines the frame's **destination MAC address**:

66:77:88:99:AA:BB

Step 3: Search the MAC address table

The switch searches its MAC address table for this destination MAC.

It finds:

66:77:88:99:AA:BB → Port 2

Step 4: Forward the frame

Because the destination MAC address is already known, the switch forwards the frame **only through Port 2**.

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Port 1  —→  Switch  —→  Port 2
Source                                Destination
```

Step 5: No flooding

The switch **does not send the frame to other ports** because it already knows where the destination device is located.

Final Answer

The switch receives the frame on **Port 1**, checks the destination MAC address, finds 66:77:88:99:AA:BB associated with **Port 2** in its MAC address table, and forwards the frame **only to Port 2**. This is called **unicast forwarding**.